

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – TECHNICAL PRESENTATION

Course Code:	ITA03	Course Title:	Mobile computing
CO Assessed:	CO3 – Precision (BL3)	Assessment:	Assessment Tool 2 – Technical Presentation
Weightage:	30 % of CIA	Total Marks:	20 (Answer ANY ONE)
CO3 Statement:	Implement wireless networking and Mobile IP concepts for routing in cellular and ad hoc networks.. (BTL 3)		
Student Name:	M.Manasvi	Reg. No.:	192311299
Section / Batch:	Slot-A	Date:	7-8-2026

Code of Conduct

I M.Manasvi -192311299 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

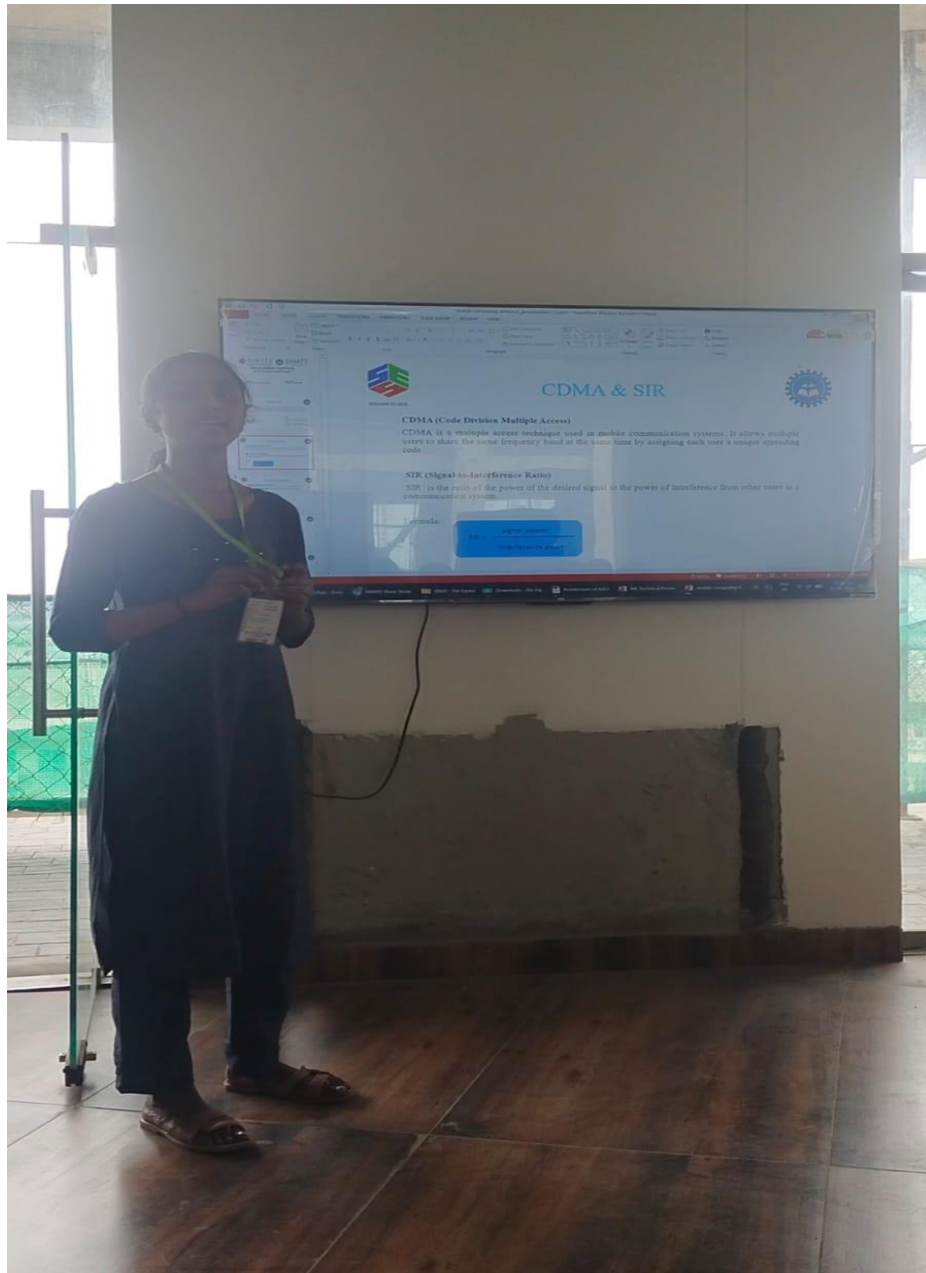
Signature of the Student:M.Manasvi

Instructions

- Duration: 10 minutes

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

1. Derive the maximum number of CDMA users supportable for a processing gain of 180 and a required linear SIR of 7, then derive the percentage change in maximum users if the required SIR rises to 10, showing both calculations. Discuss why raising the SIR requirement (e.g., to improve call quality) necessarily reduces capacity, and describe this as a defining design trade-off CDMA operators must manage continuously.





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ITA0301 MOBILE COMPUTING

CO3 AT2 TECHNICAL PRESENTATION

Guided by:
Dr. Terrance Federick Fernandez

Presented by:
M.Manasvi 192311299



PROBLEM



Derive the maximum number of CDMA users supportable for a processing gain of 180 and a required linear SIR of 7, then derive the percentage change in maximum users if the required SIR rises to 10, showing both calculations. Discuss why raising the SIR requirement (e.g., to improve call quality) necessarily reduces capacity, and describe this as a defining design trade-off CDMA operators must manage continuously.

GIVEN

A CDMA system has:

- Processing Gain (PG) = **180**
- Required Linear SIR = **7**

FIND

- Maximum number of CDMA users.
- New capacity when SIR increases to **10**.
- Percentage reduction in capacity.
- Explain why increasing SIR reduces capacity.

CDMA & SIR

CDMA (Code Division Multiple Access)

CDMA is a multiple access technique used in mobile communication systems. It allows multiple users to share the same frequency band at the same time by assigning each user a unique spreading code

SIR (Signal-to-Interference Ratio)

SIR is the ratio of the power of the desired signal to the power of interference from other users in a communication system.

Formula:

$$\text{SIR} = \frac{\text{signal power}}{\text{Interference power}}$$

Concept & Formula

The capacity of a CDMA system depends on:

- **Processing Gain (PG):** Measures how much the signal is spread, helping to reduce interference.
- **Signal-to-Interference Ratio (SIR):** Indicates the strength of the desired signal compared to interference from other users.

$$N = \frac{\text{Processing Gain}}{\text{Required SIR}}$$

- N = Maximum number of users
- PG = Processing Gain
- SIR = Required Signal-to-Interference Ratio

Calculation

➤ Calculation (SIR = 7)

PG = 180
SIR = 7

- $N = 180/7 = 25.71$
- $N \sim 26$ users

➤ Calculation (SIR = 10)

PG = 180
SIR = 10

- $N = 180/10 = 18$
- $N = 18$ users

➤ Percentage decrease

- $(26-18/26)*100 = 30.77\%$
- Capacity decreases by approximately 31%

Why Does Capacity Reduce?

Explanation

- A higher SIR requires a stronger and cleaner signal.
- To achieve this, interference from other users must be reduced.
- Since each user contributes interference, the network can support fewer users.
- Therefore, improving call quality by increasing SIR reduces the number of users the system can accommodate.



Conclusion



Results

- Processing Gain = 180
- SIR = 7 $\rightarrow$ 26 users
- SIR = 10 $\rightarrow$ 18 users
- Capacity decreases by approximately 31%

CDMA operators must balance:

- Higher SIR $\rightarrow$ Better call quality, fewer users
- Lower SIR $\rightarrow$ More users, lower call quality

Conclusion: Increasing the required SIR improves communication quality but reduces system capacity, making the balance between **quality and capacity** a key design trade-off in CDMA systems.